

Roseth Perera

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EXPERIENCE

Software Development Engineer Intern

May 2026 – Aug 2026

Amazon

Seattle, WA

- Built *Compliance-IQ*, an AI agent on AWS Bedrock AgentCore replacing manual UI navigation with natural-language access to enterprise compliance data
- Converted 16 stakeholder requirements into a prioritized backlog, shipped a 7-feature sprint against that feedback in 3 days, and demoed live to org leadership
- Root-caused a live production outage across IAM policy, Java stream parsing, and Lambda deploy versioning
- Benchmarked 3 data-layer candidates at 5–26s against a 3s p90 budget, migrated search to OpenSearch Serverless, and cut infrastructure cost 99% versus the baseline design
- Implemented fail-closed grounding with Bedrock Guardrails, a refusal-first system prompt, and prompt-injection hardening so the agent cites sources or declines rather than fabricating

Deck Supervisor / Lifeguard

Mar 2024 – Aug 2025

Goldfish Swim School

Park Ridge, IL

- Coordinated a team of lifeguards across daily shifts, communicating real-time safety decisions to staff and parents under time pressure

Volunteer Project Leader

Nov 2021 – Sep 2023

Teens Who Care Sri Lanka

Colombo, Sri Lanka

- Led student teams raising funds and delivering school supplies to rural schools during Sri Lanka's economic crisis

PROJECTS

Aeolus

Jun 2026 – Present

- Built and shipped a full-stack AI trip planner (Next.js, Supabase, Vercel) to production, grounding recommendations in live NWS/NPS data via tool-calling
- Fixed a relevance bug in the NPS API that buried the correct park behind 455 keyword matches, building a local re-ranking layer so the right park surfaces first
- Designed a server-side citation system attaching source metadata only from tools that actually succeeded, plus a per-user preference schema promoting inferred signals to fact after 3 consistent observations
- Root-caused a 4-layer serverless deployment failure (Chromium version drift, two dependency-tracing gaps, a shared-library path conflict) by reading vendored package source

Carbon-Aware AI Scheduler

Feb 2026 – Apr 2026

- Built a Python CLI scheduler ranking lower-emission execution windows for AI workloads from live grid carbon-intensity forecasts, reporting projected emissions savings per schedule from `nvidia-smi` GPU energy profiles

Battery Telemetry: State-of-Charge Tracking

Sep 2025 – May 2026

- Built a 24V LiFePO₄ telemetry system streaming ADC-sampled voltage/current to an NVIDIA Jetson over UART, implementing real-time state-of-charge estimation

EDUCATION

University of Wisconsin–Madison

Madison, WI

B.S. Electrical Engineering, Minor in Engineering for Energy Sustainability

Expected May 2029

LANGUAGES & TECHNOLOGIES

Programming: Python, Java, TypeScript, SQL

Cloud & Infra: AWS (Bedrock, Lambda, DynamoDB, OpenSearch, CDK), Azure OpenAI, Supabase/Postgres, Vercel, Docker, CI/CD, Linux

AI/ML: Retrieval-augmented generation (RAG), LLM tool-calling (MCP), embeddings

ACTIVITIES

Wisconsin Robotics (Electrical Team) | Engineers Without Borders